Software Engineering Project

DispatchNet:

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Use Case Analysis

Contents

RE: Purpose **3**

RE: Objectives **4**
0.1 Ticketing system 4
0.2 Route-finding 4
0.3 Virtual board 5
0.4 Rail system modeling 5

RE: Stakeholders **6**
Internal 6
Human 6
Human 6
External 6
Human 6
Non-human 6

1 Component Diagram **7**
1.1 User authentication 8
1.1.1 Registration 9
1.2 User management 9
1.3 Ticketing management 11
1.4 Path finding 13
1.5 Info view 15
1.6 Network management 16
1.7 Service management 17

Context Description (D1) **18**

Readability notes

Bold words are system-important constructs.
Underlined nouns are actors of the system.
Italic words are special keywords to denote important limits of the project.
Hyperlinks in reference to sections and subsections will be blue.
Hyperlinks to external web content will be in magenta.

RE: Purpose

This document describes the use cases, their relation, and the class-based diagram of Project "TRE", which implements a rail transport management system.

The purpose of this document is to:

- Represent all intended use cases
- Relate use cases with eachother and with stakeholders
- Itemize and relate the major data-components employed by the system
- Provide a top-level view of the system's operations

RE: Objectives

A major European rail infrastructure manager received an idea for a unified rail service scheduling and ticketing system, and has chosen us to prototype a trial run, over multiple regions in their network.

Thus, the project provides a platform that covers the needs of both passengers and train companies, as well as an underlying management system of any regional-sized rail network, which supports network managers in its administration.
It provides a set of functionalities which support the planning of train travel. The system lets passengers discover paths, view scheduling and manage tickets. On the other side, it lets train companies introduce train services and network managers administer the network.
The project coordinates the functionalities for each stakeholder into a solution that has been identified in an application accessible for the aforesaid users.

The main objectives of the project are:

0.1 Ticketing system

Subsystem which deals with tickets, from the search to the possible actions after the purchase. It includes basic operations that the passenger can perform, such as:

- Viewing tickets details.
- Purchase.
- Accessing tickets history.
- Cancellation.

0.2 Route-finding

The system shall let passengers select a starting station, a destination station, and either a departure time or arrival time.
It computes and ranks possible **routes** that best fit the user's request and

permissions (for example, not permitting cargo trains to passengers).
The user must be able to compare the options based on:

- Departure and arrival times.
- Number of transfers.
- Possible transfer locations.

0.3 Virtual board

The system provides passengers access to departure and arrival boards for any station. It also shows intermediate stations and scheduled times for each individual train service.

0.4 Rail system modeling

The system provides a data-driven approach to rail network modelling, supporting an efficient administration of the infrastructure.
It shall support:

- Management of the network by the network manager.
- Management of rolling stock by the network manager.
- Submission of a new service request by train companies.
- Review of a service request by the network manager.

RE: Stakeholders

Internal

Human

Network managers:, the product owner. They manage the rail network combining the availability of train companies with passengers' needs.
They are the primary stakeholders for making functional decisions.

Non-Human

Database, where all the information used by the application are stored.

External

Human

Passengers, the main end users. They want to streamline their interactions with the train transport system, including:

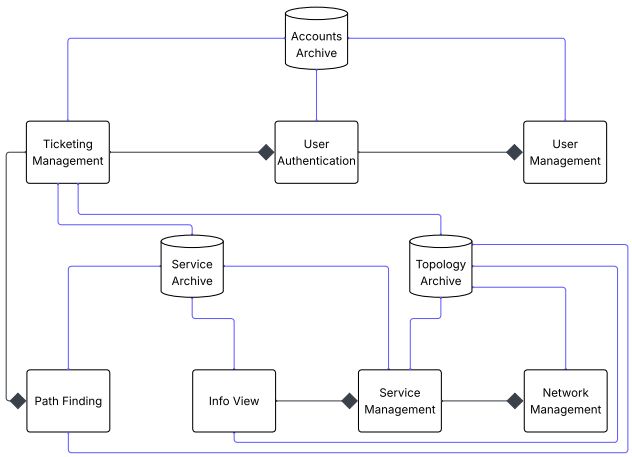
- Searching for routes and schedules.
- Analyzing departure/arrival boards as well as train information.
- Purchasing tickets.

Train companies, who provide the rolling stock. They request the introduction of services. They are also expected to provide information of trains they ask to be managed.
Anonymous users, who have not yet authenticated. They are considered as passengers with restricted access to functionalities that require authentication.

Non-human

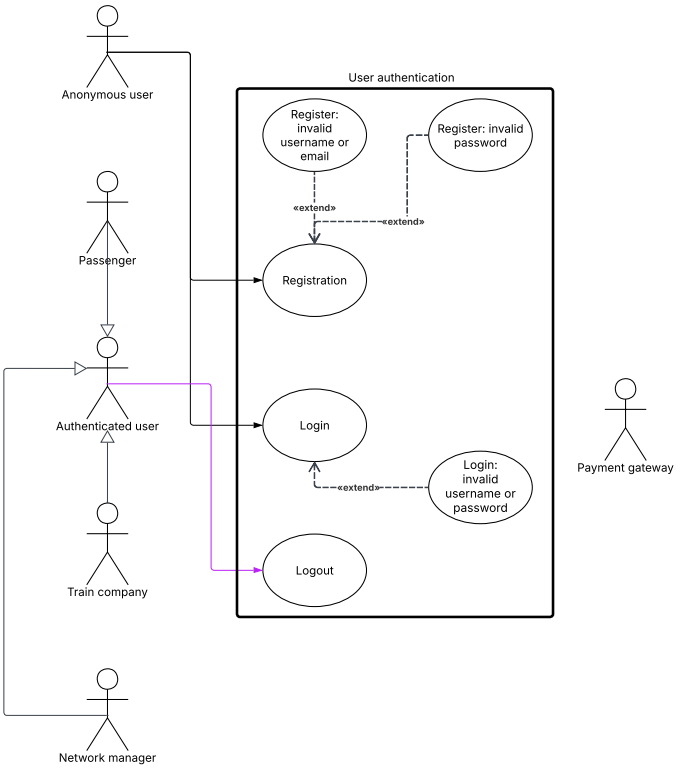
Payment gateway, which provides the underlying payment methods.

1 Component Diagram



Alternate link at our Github

1.1 User authentication



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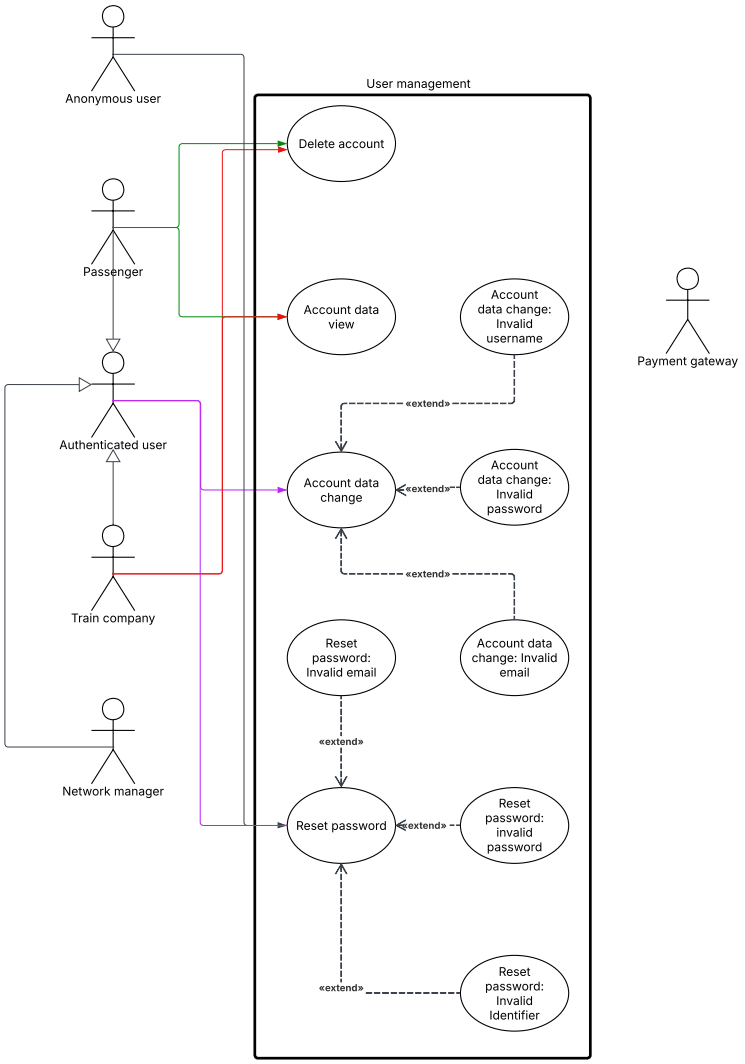
1.1.1 Registration

1.2 User management



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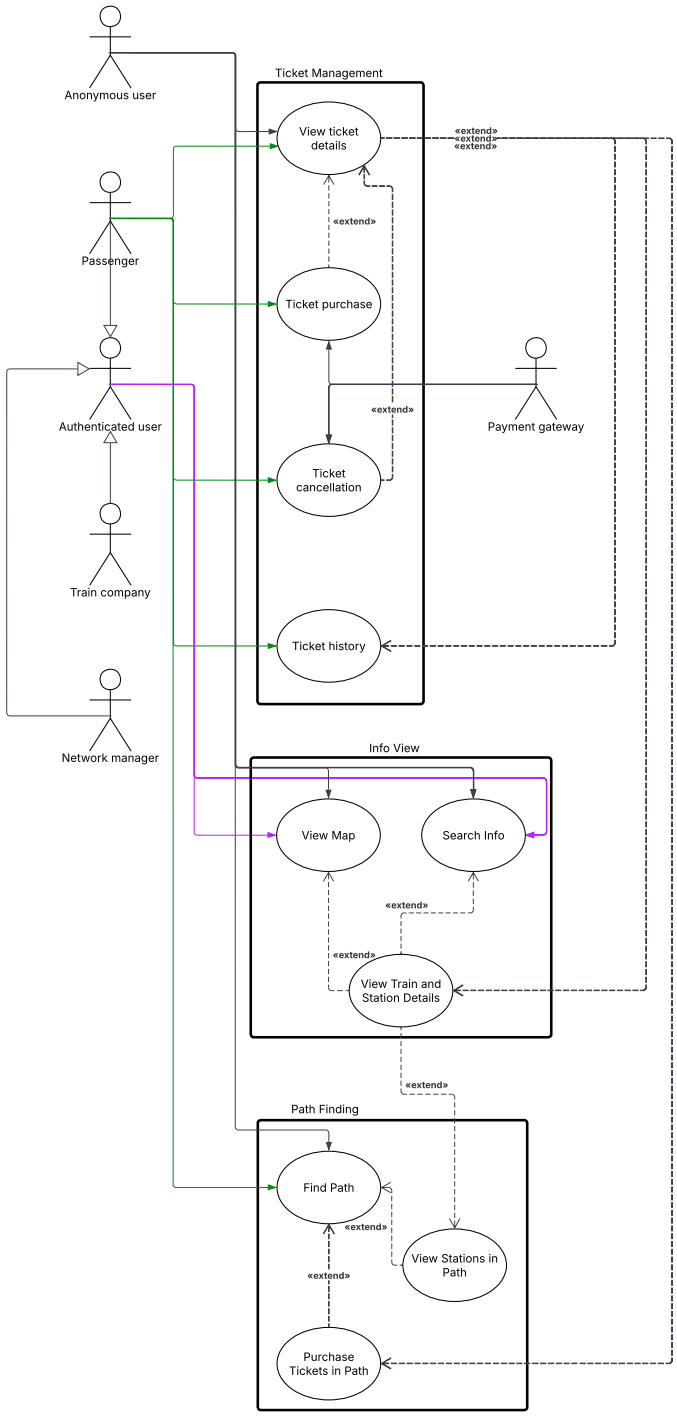
1.3 Ticketing management



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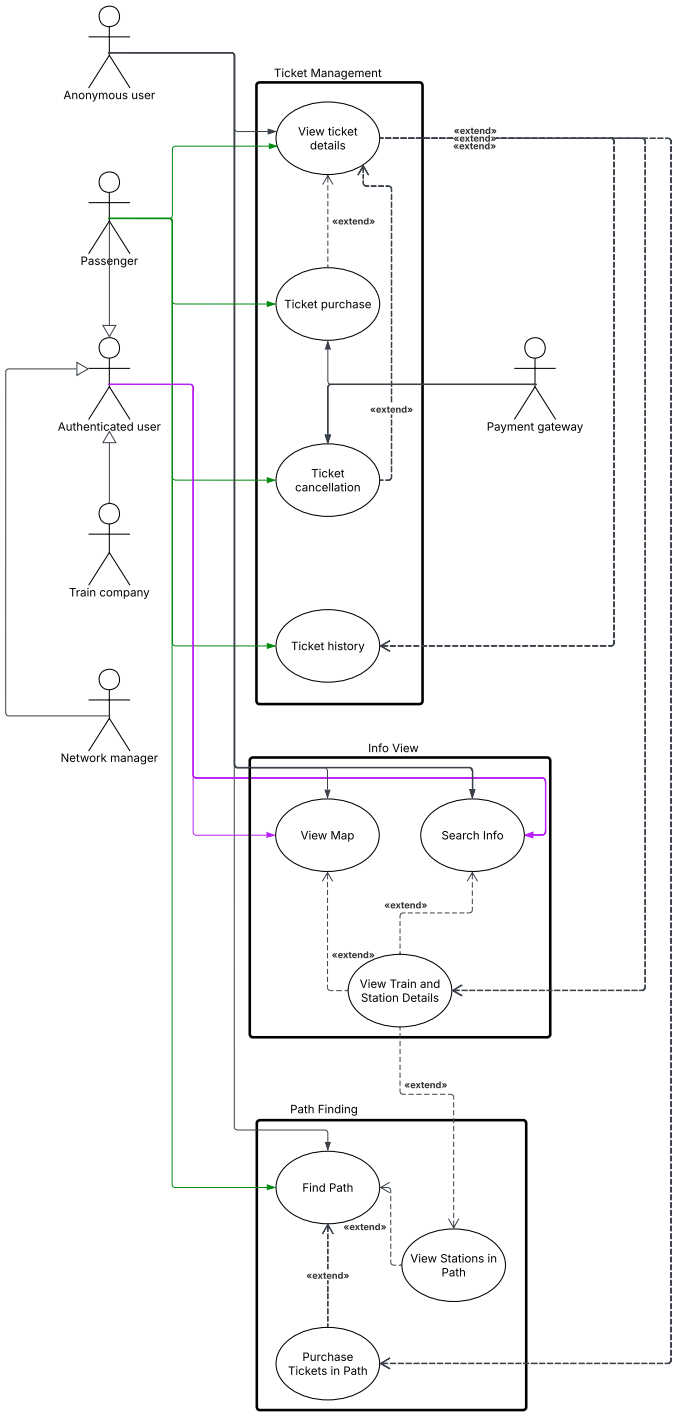
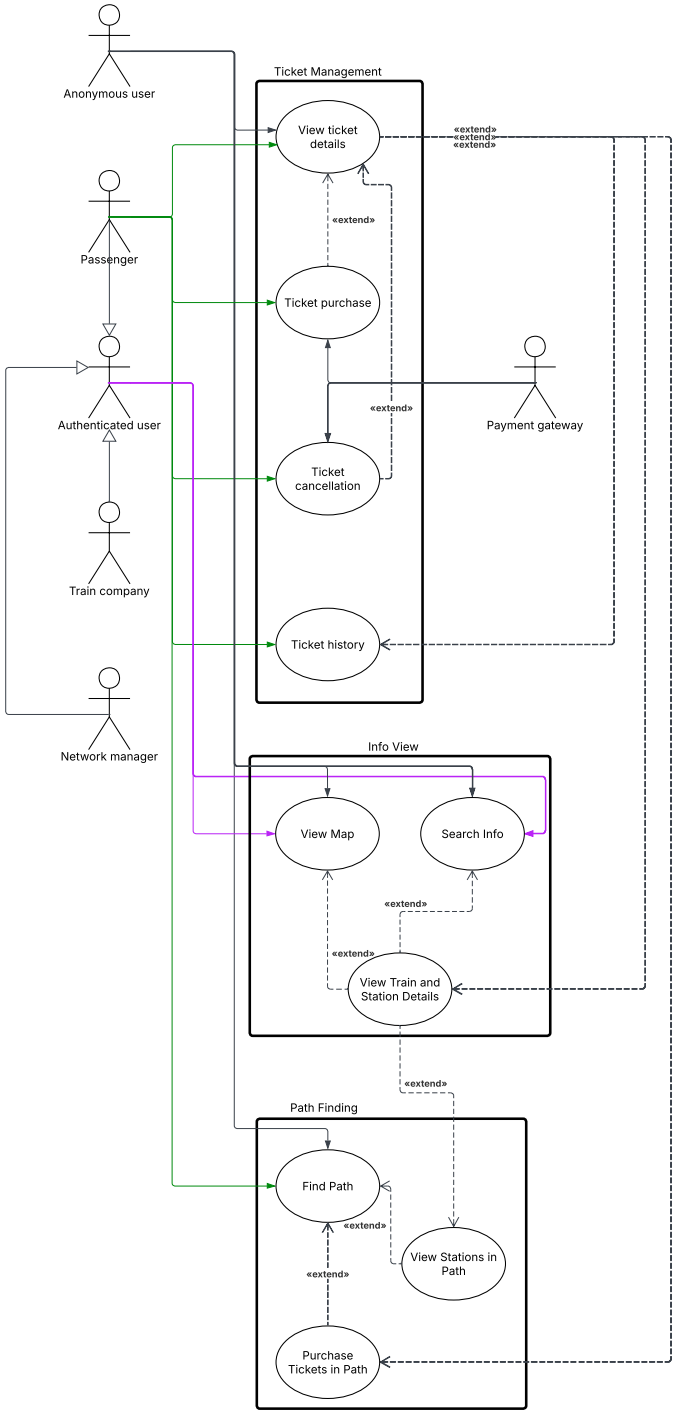
1.4 Path finding



Alternate link at our Github

1.5 Info view



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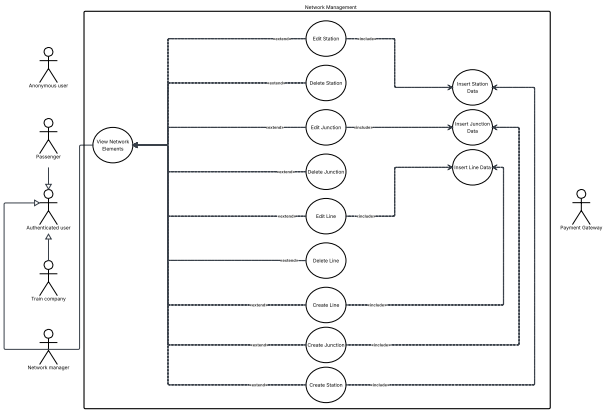
1.6 Network management



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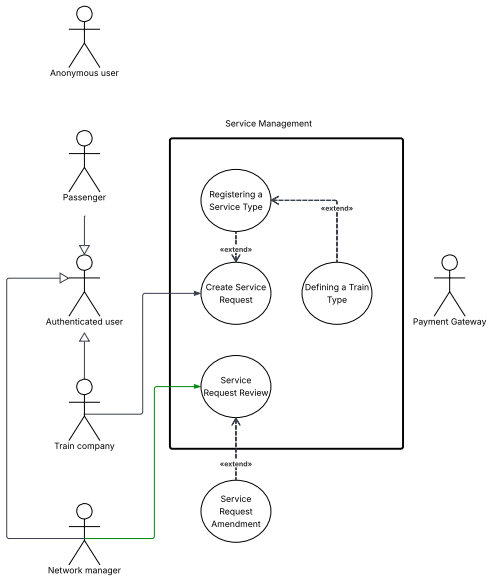
1.7 Service management



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Requirements Analysis

Contents

Purpose **5**

O Objectives **5**
O.1 Ticketing system 5
O.2 Route-finding 6
O.3 Virtual board 6
O.4 Rail system modeling 6

Stakeholders **6**
Internal 7
Human 7
Human 7
External 7
Human 7
Non-human 7

FR Functional requirements **8**
FR.1 Account classification 8
FR.2 Account registration 8
FR.2.1 Consent management 8
FR.2.2 Account registration notification 8
FR.3 Account deletion 8
FR.3.1 Account deletion notification 8
FR.4 Data request management 8
FR.5 Account data change 9
FR.5.1 Account data change notification 9
FR.6 Password reset 9
FR.7 Login 9
FR.8 Logout 9
FR.9 Tickets details display 9
FR.10 Tickets purchase 9
FR.10.1 Tickets purchase confirmation 10
FR.10.2 Tickets purchase notification 10
FR.11 Tickets cancellation 10
FR.11.1 Tickets cancellation confirmation 10
FR.11.2 Tickets cancellation notification 10

FR.12 Tickets history 10
FR.12.1 Tickets classification 10
FR.13 Path search 10
FR.13.1 Path points 11
FR.13.2 Path timing 11
FR.14 Path ranking 11
FR.14.1 Ranking length 11
FR.14.2 Ranking deviation "Departure" 11
FR.14.3 Ranking deviation 2 "Arrival" 11
FR.14.4 Ranking cost 11
FR.15 Transfers 12
FR.16 Train access restriction 12
FR.17 Station display 12
FR.17.1 Station search 12
FR.18 Train display 12
FR.18.1 Train search 12
FR.19 Path display 12
FR.20 Creating a Service Request 13
FR.21 Defining a Train Type 13
FR.22 Defining a Service Type 13
FR.23 Viewing the Network 13
FR.24 Managing a Station 13
FR.24.1 Registering a Station 13
FR.24.2 Modifying a Station 13
FR.24.3 Deleting a Station 13
FR.25 Managing a Line 14
FR.25.1 Registering a Line 14
FR.25.2 Modifying a Line 14
FR.25.3 Deleting a Line 14
FR.26 Managing a Junction 14
FR.26.1 Registering a Junction 14
FR.26.2 Modifying a Junction 14
FR.26.3 Deleting a Junction 14
FR.27 Automated Service Integrity Checks 15
FR.27.1 Single Track Line Availability Check 15
FR.27.2 Platform Availability Check 15
FR.27.3 Track-side Systems Compatibility Check 15
FR.27.4 Track Geometry Check 15

FR.28 Displaying Integrity Check Results 15
FR.29 Service Request Review 15
FR.29.1 Service Request approval 16
FR.29.2 Service Request amendment 16
FR.29.3 Service Request denial 16
Requirement restrictions based on user classification 16

NFR Non-functional requirements 17
NFR.1 Data Protection Regulation 17
NFR.2 Account data 17
NFR.3 Account data details 18
NFR.4 Password security 18
NFR.5 Typo prevention 18
NFR.6 Consent for personal data processing 18
NFR.7 Request of processed data 18
NFR.8 Rectification of processed data 19
NFR.9 Email notification 19
NFR.10 Station-train display switching 19
NFR.10.1 Station to train display 19
NFR.10.2 Train to station display 19
NFR.11 Station-path searching switching 19
NFR.11.1 Station "from" path 19
NFR.11.2 Station "to" path 20
NFR.12 Path list to path list 20
NFR.13 Ranking non-choice 20
NFR.14 Network infrastructure 20

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Green text is text added or modified for D2.

Purpose

This document describes the scope and the requirements of the Project "TRE", which implements a rail transport management system.

The purpose of this document is to:

- Specify the objectives.
- Identify the primary stakeholders.
- Define the functional and non-functional requirements.

O Objectives

A major European rail infrastructure manager received an idea for a unified rail service scheduling and ticketing system, and has chosen us to prototype a trial run, over multiple regions in their network.

Thus, the project provides a platform that covers the needs of both passengers and train companies, as well as an underlying management system of any regional-sized rail network, which supports network managers in its administration.
It provides a set of functionalities which support the planning of train travel. The system lets passengers discover paths, view scheduling and manage tickets. On the other side, it lets train companies introduce train services and network managers administer the network.
The project coordinates the functionalities for each stakeholder into a solution that has been identified in an application accessible for the aforesaid users.

The main objectives of the project are:

O.1 Ticketing system

Subsystem which deals with tickets, from the search to the possible actions after the purchase. It includes basic operations that the passenger can perform, such as:

- Viewing tickets details.

- Purchase.
- Accessing tickets history.
- Cancellation.

O.2 Route-finding

The system shall let passengers select a starting station, a destination station, and either a departure time or arrival time.
It computes and ranks possible **routes** that best fit the user's request and permissions (for example, not permitting cargo trains to passengers).
The user must be able to compare the options based on:

- Departure and arrival times.
- Number of transfers.
- Possible transfer locations.

O.3 Virtual board

The system provides passengers access to departure and arrival boards for any station. It also shows intermediate stations and scheduled times for each individual train service.

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The system provides a data-driven approach to rail network modelling, supporting an efficient administration of the infrastructure.
It shall support:

- Management of the network by the network manager.
- Management of rolling stock by the network manager.
- Submission of a new service request by train companies.
- Review of a service request by the network manager.

Stakeholders

Internal

Human

Network managers, the product owner. They manage the rail network combining the availability of train companies with passengers' needs.
They are the primary stakeholders for making functional decisions.

Non-Human

Database, where all the information used by the application are stored.

External

Human

Passengers, the main end users. They want to streamline their interactions with the train transport system, including:

- Searching for routes and schedules.
- Analyzing departure/arrival boards as well as train information.
- Purchasing tickets.

Train companies, who provide the rolling stock. They request the introduction of services. They are also expected to provide information of trains they ask to be managed.
Anonymous users, who have not yet authenticated. They are considered as passengers with restricted access to functionalities that require authentication.

Non-human

Payment gateway, which provides the underlying payment methods.

FR Functional requirements

FR.1 Account classification

The system shall provide different functionalities, distinguishing users between between anonymous user, passenger, train company operator and network manager. The additional functionalities are described in Requirement restrictions based on user classification.

FR.2 Account registration

Following FR.1, in reference to NFR.2, the system must allow anonymous users to register. Data must satisfy NFR.3. Password must satisfy NFR.4 and NFR.5.

FR.2.1 Consent management

Following NFR.6, in reference to FR.2, at registration the system must request the user's consent for personal data processing.

FR.2.2 Account registration notification

In reference to FR.2, after successful registration the system must notify the user as described at NFR.9.

FR.3 Account deletion

Following NFR.1, the system shall allow users to delete their accounts. Any other related data is also deleted.

FR.3.1 Account deletion notification

In reference to FR.3, after successful deletion the user is notified as described at NFR.9.

FR.4 Data request management

Following NFR.7, in reference to NFR.2, the system must provide users view of their stored information.

FR.5 Account data change

Following NFR.8, in reference to NFR.2, the system must allow users to change their data. New data must satisfy NFR.3.

FR.5.1 Account data change notification

In reference to FR.5, after a successful change operation the user is notified as described at NFR.9.

FR.6 Password reset

Following NFR.8, in reference to NFR.2, the system shall allow users to reset the password and change it (NFR.9). New password must satisfy NFR.4 and NFR.5.

FR.7 Login

Following FR.1, the system must let users login using their username and password defined at registration (FR.2).

Boundary: The network manager has default account username "admin" and password "Password1234".

FR.8 Logout

Following FR.1, the system shall let logged in users log out, switching to the restricted anonymous user's functionalities.

FR.9 Tickets details display

Following O.1, the system must allow passengers to view tickets.

FR.10 Tickets purchase

Following O.1, the system must allow passengers to buy tickets.

FR.10.1 Tickets purchase confirmation

Following O.1, in reference to FR.10, the passenger must be asked for a final confirmation before purchase.

FR.10.2 Tickets purchase notification

Following O.1, in reference to FR.10, at the end of a purchase operation, the system must notify the passenger as described at NFR.9.

FR.11 Tickets cancellation

Following O.1, the system must allow the cancellation of purchased tickets.

FR.11.1 Tickets cancellation confirmation

Following O.1, in reference to FR.11 and NFR.9, the passenger must be asked for a final confirmation before cancellation.

FR.11.2 Tickets cancellation notification

Following O.1, in reference to FR.11, at the end of a cancellation operation, the system must notify the passenger as described at NFR.9.

FR.12 Tickets history

Following O.1, the system must allow passengers to view their tickets history, both purchased and cancelled.

FR.12.1 Tickets classification

Following O.1, in reference to FR.12, the tickets history must show whether a ticket is valid, expired, or cancelled.

FR.13 Path search

Following O.2, the system needs to be able to compute multiple paths between stations , that is an ordered sequence of services from a departure to an arrival station.

FR.13.1 Path points

Following O.2, extending FR.13, the passenger must be able to choose both the first and last stations of paths to be computed.

FR.13.2 Path timing

Following O.2, extending FR.13, the passenger shall be able to optionally specify a departure time and/or a latest arrival time. If neither is specified, the system must consider paths with departure times starting from the current time.

FR.14 Path ranking

Following O.2, in reference to FR.13, the system needs to rank the paths with a criteria selected by the passenger among a fixed list.

FR.14.1 Ranking length

Following O.2, in reference to FR.14, the system shall rank paths by prioritizing those with shortest travel time, and, secondarily, those with fewer intermediate stations.

FR.14.2 Ranking deviation "Departure"

Following O.2, in reference to FR.14, the system shall rank paths by prioritizing those whose first departure time is closest (but not earlier than) the user requested departure time.

FR.14.3 Ranking deviation 2 "Arrival"

Following O.2, in reference to FR.14, the system shall rank paths by prioritizing those whose last arrival time is closest (but not later than) the user requested departure time.

FR.14.4 Ranking cost

Following O.2, in reference to FR.14, the system shall rank paths by prioritizing those with lower total cost of tickets not already owned by the user.

FR.15 Transfers

Following O.2, in reference to FR.13, the system needs to compute transfer paths out of intersecting routes.

FR.16 Train access restriction

Following O.2, in reference to FR.13, the system must not consider train routes that pass by trains that the passenger is not allowed to purchase a ticket for, such as cargo trains or trains with unavailable seats.

FR.17 Station display

Following O.3, the system must provide departure and arrival times of all trains that pass through any station.

FR.17.1 Station search

Following O.3, the system must allow users to search such information by a station's name.

FR.18 Train display

Following O.3, the system must provide departure and arrival times of a train at all the stations on its schedule.

FR.18.1 Train search

Following O.3, the system must allow user to search such information by the a train's identifier.

FR.19 Path display

The system must provide the board described at FR.18 for each train in a given path, limited to the segments in which the train participates.

FR.20 Creating a Service Request

Following O.4, companies must be allowed to send the system a request for the introduction of a service.

FR.21 Defining a Train Type

Following O.4, in reference to FR.22, the system must allow companies to register a train type to be used when defining a service.

FR.22 Defining a Service Type

Following O.4, in reference to FR.20 and FR.21. The system must allow companies to define services and classes, as well as a pricing model for passenger trains.

FR.23 Viewing the Network

Following O.3, the system must allow users to access a map of the network.

FR.24 Managing a Station

Following O.4, in reference to NFR.14, the system must allow network managers to manage a station.

FR.24.1 Registering a Station

Following O.4, in reference to FR.24, the system must allow network managers to register a station.

FR.24.2 Modifying a Station

Following O.4, in reference to FR.24, the system must allow network managers to modify any existing station.

FR.24.3 Deleting a Station

Following O.4, in reference to FR.24, the system must allow network managers to delete any existing station.

FR.25 Managing a Line

Following O.4, in reference to NFR.14, the system must allow network managers to manage a line.

FR.25.1 Registering a Line

Following O.4, in reference to FR.25, the system must allow network managers to register a line.

FR.25.2 Modifying a Line

Following O.4, in reference to FR.25, the system must allow network managers to modify any existing line.

FR.25.3 Deleting a Line

Following O.4, in reference to FR.25, the system must allow network managers to delete any existing line.

FR.26 Managing a Junction

Following O.4, in reference to NFR.14, the system must allow network managers to manage a junction.

FR.26.1 Registering a Junction

Following O.4, in reference to FR.26, the system must allow network managers to register a junction.

FR.26.2 Modifying a Junction

Following O.4, in reference to FR.26, the system must allow network managers to modify any existing junction.

FR.26.3 Deleting a Junction

Following O.4, in reference to FR.26, the system must allow network managers to delete any existing junction.

FR.27 Automated Service Integrity Checks

Following O.4, the system must autonomously perform integrity checks on proposed services. Integrity checks are defined by the following.

FR.27.1 Single Track Line Availability Check

Following O.4, specifying FR.27, the system must ensure that no two services are scheduled to traverse a single track line in opposite directions at the same time.

FR.27.2 Platform Availability Check

Following O.4, specifying FR.27, the system must ensure that no two services are scheduled to stop at the same platform at the same time.

FR.27.3 Track-side Systems Compatibility Check

Following O.4, specifying FR.27, the system must ensure that all trains used for a service are compatible with the signaling and electrification systems installed on the traversed infrastructure.

FR.27.4 Track Geometry Check

Following O.4, specifying FR.27, the system must ensure that all trains used for a service are compatible with both the loading and track gauges of the infrastructure they traverse.

FR.28 Displaying Integrity Check Results

Following O.4, extending FR.27, the system must provide the integrity check results.

FR.29 Service Request Review

Following O.4, in reference to FR.20, the system must allow the network manager to review requests.

FR.29.1 Service Request approval

Following O.4, in reference to FR.20 and FR.29, the network manager must be able to approve a service request, notifying (NFR.9) the train company that submitted the request.

FR.29.2 Service Request amendment

Following O.4, in reference to FR.20 and FR.29, the system must allow the network manager to send back an amended version of the request for the train company to review, notifying (NFR.9) them.
Boundary: the system will not implement any amendment functionalities.

FR.29.3 Service Request denial

Following O.4, in reference to FR.20 and FR.29, the system must allow the network manager to deny a service request, notifying (NFR.9) the train company that submitted the request.

Requirement restrictions based on user classification

Requirement	Anonymous	Passenger	Train company	Manager
FR.1 **FR.6** **FR.23**	Yes	Yes	Yes	Yes
FR.2 **FR.7**	Yes			
FR.3-5		Yes	Yes	
FR.8		Yes	Yes	Yes
FR.9 **FR.13-19**	Yes	Yes		
FR.10-12		Yes		
FR.20-22			Yes	
FR.24-28				Yes
FR.29			Self	Self

Caption:
Self: access to resources regarding only the personal part of the functionality.

NFR Non-functional requirements

NFR.1 Data Protection Regulation

External requirement - Legislative requirement

The system must respect the Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data. With special attention to:

Art. 5 Principles relating to processing of personal data, ensuring ethical processing.

Art. 6, 7 Lawfulness of processing and Conditions for consent, both concerning consent for the processing of personal data.

Art. 13 Information to be provided where personal data are collected from the data subject, i.e. providing information of the controller of the processing.

Art. 15 Right of access by the data subject, i.e. users must be able to see their stored information.

Art. 16 Right to rectification, i.e. users must be able to change their information.

Art. 17 Right to erasure ('right to be forgotten').

Art. 32 Security of processing.

NFR.2 Account data

Product requirement - Security requirement

Following FR.1 and NFR.1, in reference to FR.2, user accounts must include username, email and password.

NFR.3 Account data details

Product requirement - Security requirement

Following NFR.1 and NFR.2, account data must satisfy the following:

- The username must be unique.
- The email must be unique.

NFR.4 Password security

Product requirement - Security requirement

Following NFR.1 and NFR.2, in reference to FR.2.1, the password must be at least 8 characters, with at least one upper and one lowercase letters and one digit.

NFR.5 Typo prevention

Product requirement - Security requirement

When a password is chosen or modified, the system must take prevention to mitigate typos: it must require the password to be typed twice and only accepted if both match.

NFR.6 Consent for personal data processing

External requirement - Legislative requirement

Following NFR.1, the system must comply with the GDPR for data processing consent.

NFR.7 Request of processed data

External requirement - Legislative requirement

Following NFR.1, the system must comply with the GDPR for accessing processed data.

NFR.8 Rectification of processed data

External requirement - Legislative requirement

Following NFR.1, the system must comply with the GDPR for the right of rectification.

NFR.9 Email notification

Product requirement - Usability requirement

In order to notify users, the system shall use the email service, using the one inserted at registration (FR.2).

NFR.10 Station-train display switching

Product requirement - Usability requirement
Users must be able to switch with a single prompt/input between the following

NFR.10.1 Station to train display

A station's display and a train's display passing through that station.

NFR.10.2 Train to station display

A train's display and a station's display in its schedule.

NFR.11 Station-path searching switching

Product requirement - Usability requirement
Users must be able to switch with a single prompt/input between the following.

NFR.11.1 Station "from" path

A station's display and the route-finding procedure with the station automatically set as the path's start.

NFR.11.2 Station "to" path

A station's display and the route-finding procedure with the station automatically set as the destination.

NFR.12 Path list to path list

Product requirement - Usability requirement

Users must be able to switch from a list of paths according to a ranking option to another, in no more steps/inputs than the ones required to select a different ranking.

NFR.13 Ranking non-choice

Product requirement - Usability requirement

The system shall default to the ranking described at FR.14.1 until a different ranking choice is given by the user.

NFR.14 Network infrastructure

Organizational requirement - Development requirement
The system structures the network infrastructure with junctions, stations and lines; in a way similar to a graph, where junctions and stations are nodes and lines are edges.